Exercise 5 – Hugging Face and Google Colab

1. Hugging Face Space — Text-to-Image Demo

Space used: SenseNova-U1.5-8B-MoT (unified text-to-image and image editing)

Prompt used: "A futuristic city skyline at night with flying vehicles, neon lights reflecting on wet streets, cyberpunk style"

Settings: Aspect ratio 16:9, Seed 1432621567

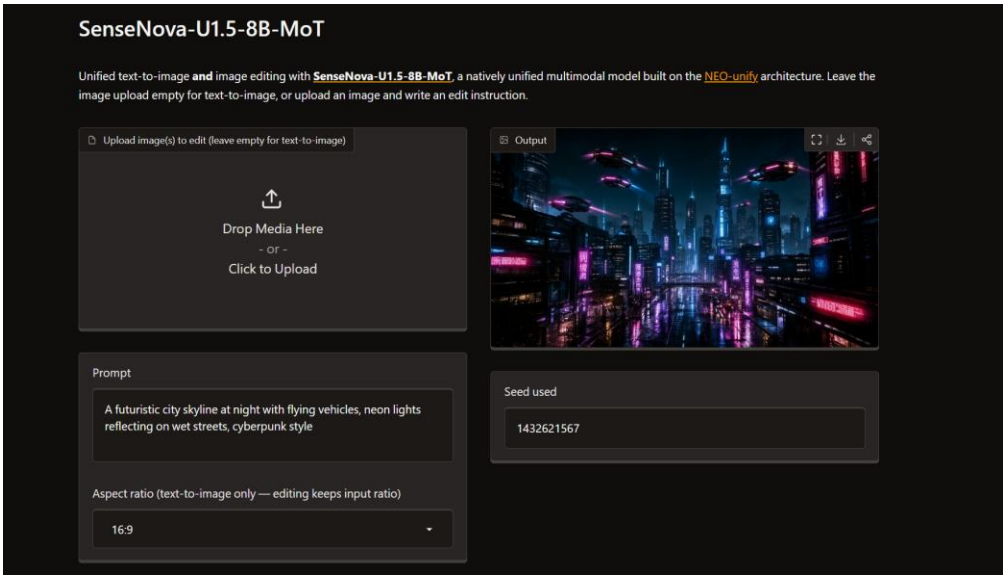

Figure 1: Generated cyberpunk city skyline using the SenseNova-U1.5-8B-MoT Space.

2. Exploring a Model on Hugging Face

Model explored: HauhauCS/Qwen3.8-27B-Uncensored-HauhauCS-Aggressive-MTP-GGUF (Image-Text-to-Text, GGUF quantized, Apache-2.0 license).

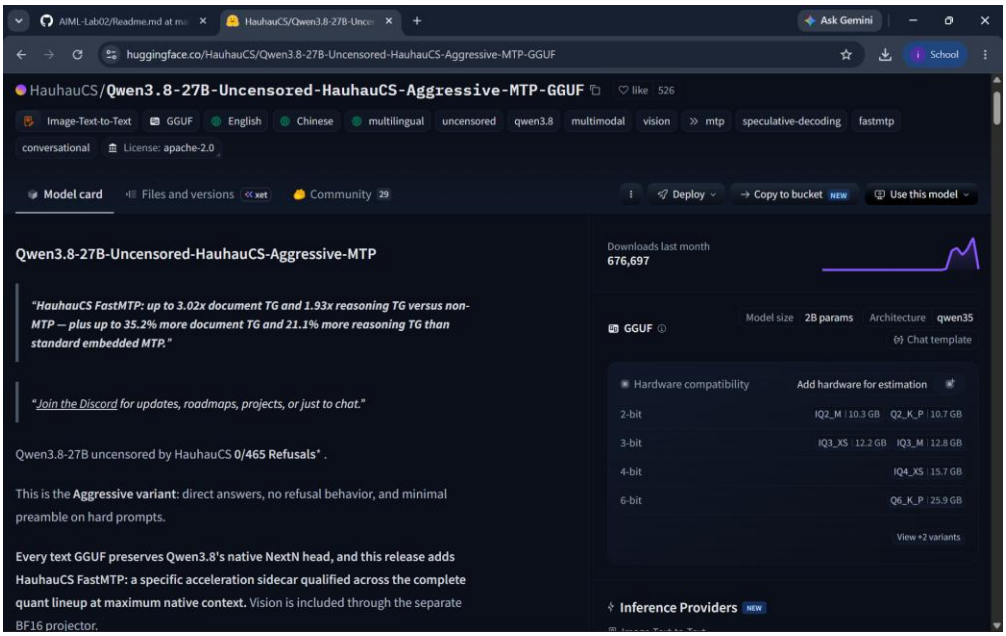

Figure 2: Model card overview, tags, and hardware compatibility table.

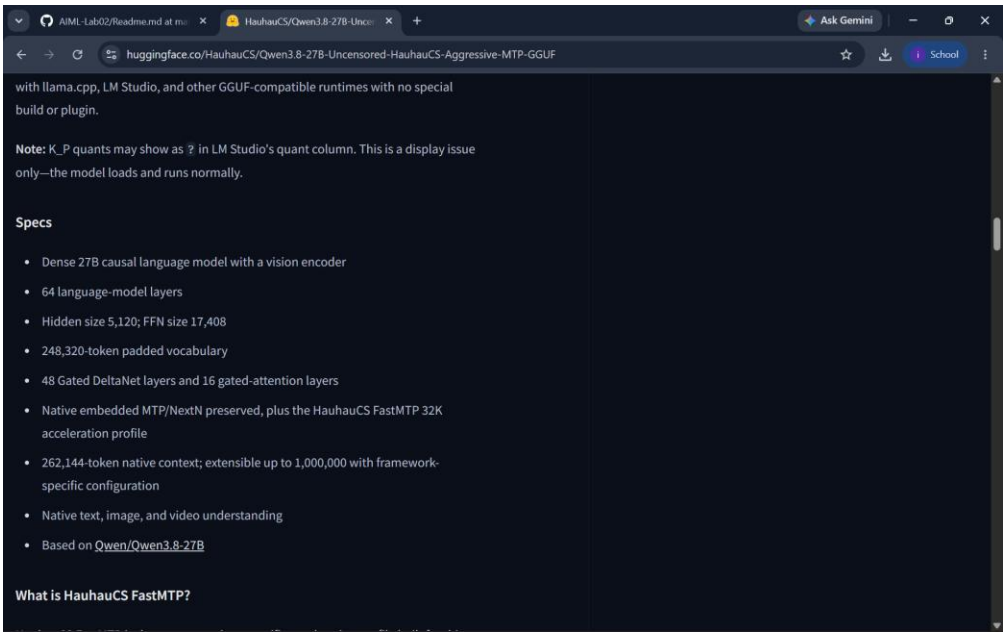

Figure 3: Model specs — 27B dense causal language model with vision encoder, 262K native context.

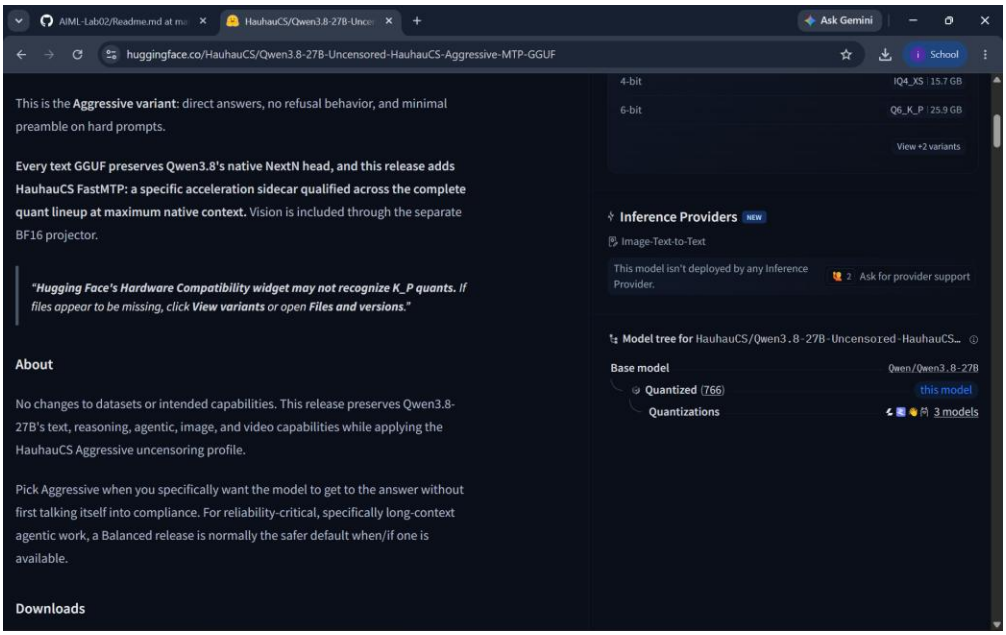

Figure 4: Model tree showing base model (Qwen3.8-27B) and quantization details.

4. Step 9 — General Summarization Pipeline

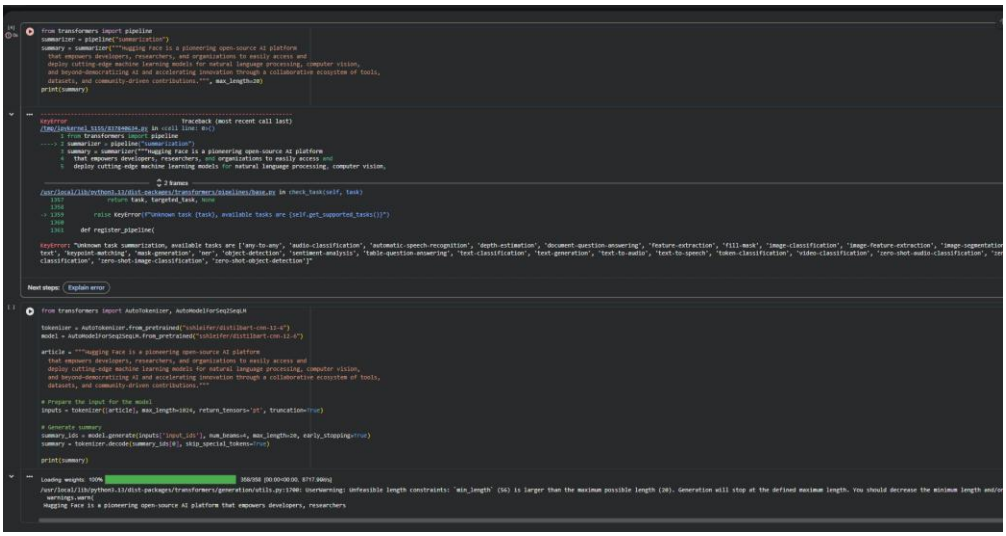

Figure 5: pipeline() KeyError (top) and the successful workaround using AutoTokenizer/AutoModelForSeq2SeqLM (bottom).

5. Step 11 — kobart-summary-v3 Model

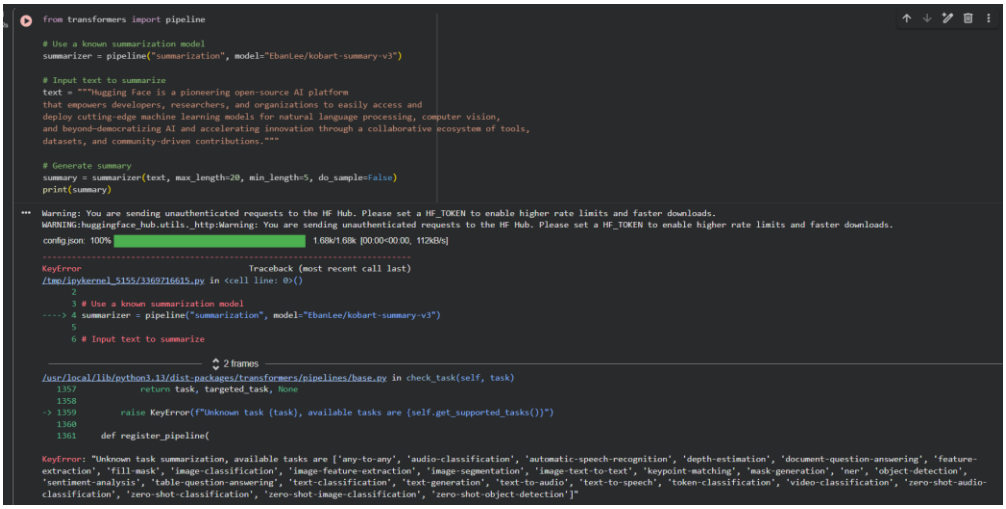

Figure 6: Same pipeline KeyError encountered when loading the kobart-summary-v3 model.

6. Step 12–13 — Model of Choice: HauhauCS Qwen3.8-27B (GGUF)

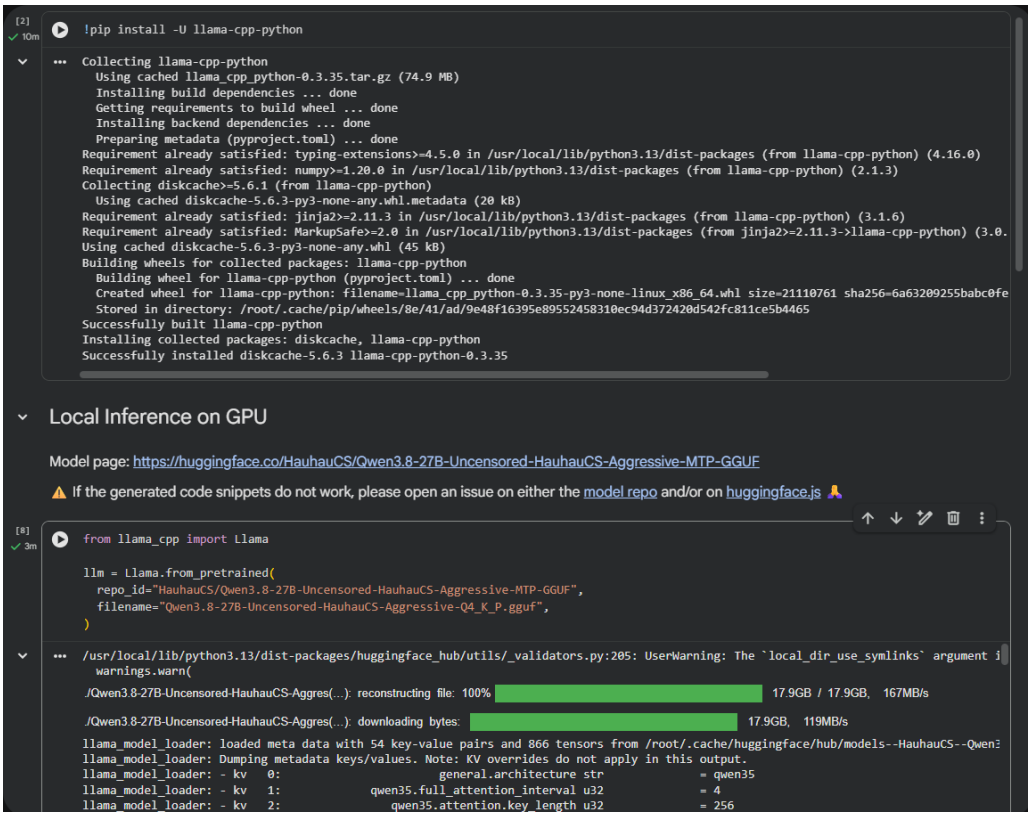

Figure 7: Auto-generated Colab code for local GPU inference with the chosen model.

7. Reflection

The main challenge in this exercise was an environment-level issue where pipeline() could not resolve the "summarization" task despite it being a standard Hugging Face pipeline type — this was worked around by loading the tokenizer and model classes directly. Exploring a large, quantized, vision-capable model (27B parameters, GGUF format) also highlighted the practical trade-offs of running large open-source models locally: multiple quantization levels are offered specifically to fit different hardware/VRAM budgets, from ~10GB (2-bit) up to ~26GB (6-bit), which directly affects whether a model can run on Colab's free-tier GPU at all.